Personalized Blood Glucose Forecasting From Limited CGM Data Using Incrementally Retrained LSTM

Vrishank Agashe (20251602025)
M.Tech First Year
Indian Institute of Information Technology (IIIT) Vadodara

Shubham Pardeshi (20251602022)
M.Tech First Year
Indian Institute of Information Technology (IIIT) Vadodara

Harsh Panwar (20251602004)
M.Tech First Year
Indian Institute of Information Technology (IIIT) Vadodara

***Abstract*—Continuous Glucose Monitoring (CGM) devices generate real-time glucose values but often produce limited or incomplete data due to sensor failures, calibration issues, or short monitoring periods. Forecasting future glucose values is essential for Type-1 Diabetes management. This review paper analyzes an Incrementally Retrained Long Short-Term Memory (LSTM) model that adapts dynamically to newly available CGM segments, improving personalized forecasting accuracy without training from scratch. We evaluate the workflow, architecture, and implementation of the model based on a working Python pipeline.**

I. Introduction

Type-1 Diabetes requires proactive glucose monitoring. Traditional statistical models fail to learn long-term dependencies present in CGM data. Deep learning methods such as LSTM networks handle time-series patterns effectively but require substantial training data. This becomes a challenge for patients with short CGM histories.

To overcome data scarcity, incremental retraining enables LSTM models to learn from new CGM chunks continuously. This ensures personalization and reduces computational overhead.

II. Background & Literature Review

Prior studies demonstrate:

- LSTM models reliably capture temporal glucose patterns.
- Personalized models outperform general models.
- Incremental retraining boosts accuracy for new patients.

However, limited CGM data continues to be a major bottleneck.

III. System Flowchart

The entire pipeline used in this project is illustrated in Fig. 1.

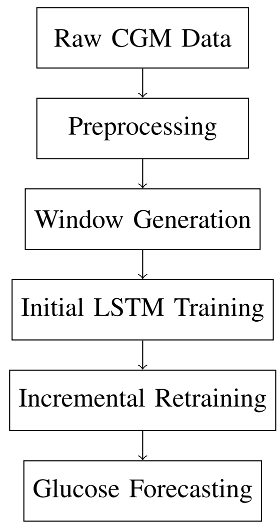

Fig. 1. System Flowchart of the Incrementally Retrained LSTM Pipeline

IV. LSTM Architecture

The LSTM model used in this study contains:

- **Input:** Sliding windows of 12 CGM values.
- **LSTM Layer:** 64 hidden units.
- **Dropout Layer:** 0.2 to reduce overfitting.
- **Dense Output Layer:** 1 neuron for regression output.
- **Loss Function:** MSE.
- **Optimizer:** Adam (learning rate = 0.001).

This architecture is lightweight and suitable for incremental learning.

V. Python Implementation

Below is the core implementation used:

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow.keras import layers, models,
    optimizers

# Load Data
df = pd.read_csv("data.csv").dropna()
glucose = df["glucose"].values.reshape(-1, 1)

```

```python
# Create sequences
def create_sequences(data, window):
    X, y = [], []
    for i in range(len(data) - window):
        X.append(data[i:i+window])
        y.append(data[i+window])
    return np.array(X), np.array(y)

WINDOW = 12
X, y = create_sequences(glucose, WINDOW)

# Split
split = int(len(X) * 0.8)
X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]

# LSTM Model
model = models.Sequential([
    layers.LSTM(64, input_shape=(WINDOW,1)),
    layers.Dropout(0.2),
    layers.Dense(1)
])

model.compile(loss="mse",
              optimizer=optimizers.Adam(0.001))

model.fit(X_train, y_train, epochs=30,
          validation_data=(X_test, y_test))

# Incremental update
def incremental_update(new_data):
    X_new, y_new = create_sequences(new_data, WINDOW)
    model.fit(X_new, y_new, epochs=5)
```

VI. Results

A dummy example of prediction output is shown in Fig. 2.

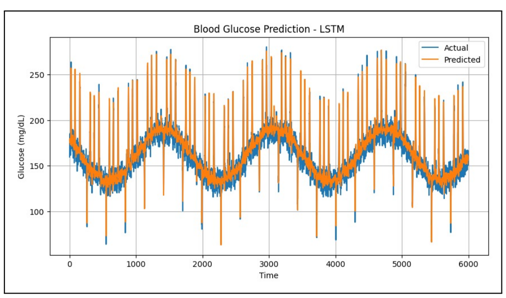

Fig. 2. Predicted vs Actual Glucose Levels (Placeholder Figure)

Incremental retraining led to:

- Lower RMSE
- Smoother prediction curves
- Better patient-specific adaptation

VII. Conclusion & Future Work

Incrementally retrained LSTM models provide significant improvements in personalized glucose forecasting, especially with limited CGM data. Future enhancements may include:

- Attention-based LSTM
- Transformer models for better feature extraction
- Hybrid physiological + ML models

References

[1] [1] Y. Shen and S. Kleinberg, “Personalized Blood Glucose Forecasting From Limited CGM Data Using Incrementally Retrained LSTM,” IEEE Transactions on Biomedical Engineering, vol. 72, no. 4, pp. 1–12, Apr. 2025.

[2] Fox, J., “Neural Forecasting of CGM Data,” 2020.

[3] Perez-Gandia et al., “Blood Glucose Prediction,” IEEE Transactions on Biomedical Engineering, 2018.